FRONT PAGE OF THE ANSWER SCRIPT

- Examination (Write the official title of the examination as it appears at the head of the question paper: ………………………………… ………………………………… …………………………………

- Title of the paper and Subject code: ………………………………… …………………………………

- Index Number (Write very clearly): …………………………………

EXAMPLE

- Examination (Write the official title of the examination as it appears at the head of the question paper: Second Year Semester II Examination

- Title of the paper and Subject code: Inorganic Chemistry, CHE 2205

- Index Number (Write very clearly): 0000

- **Use the following format to name your answer script:**
  Index #_registration# (Ex: 9999,ASB.2016.2017.000)
- **Upload your answer script to LMS**
- **Any inquiry call/ Whatsapp/ viber: 0715408587/ 0712943957/ 0715376465**
- **After submission use the comment section in LMS to clarify any issues in submission**

RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

B.Sc. in Applied Sciences
Second year Semester I Examination – January/February 2021

CHE 2205 – INORGANIC CHEMISTRY

Answer all questions **Time: Two (2) hours.**

Mass of electron	m_e	=	9.11×10^{-31} kg
Mass of proton	m_p	=	1.672×10^{-27} kg
Mass of neutron	m_n	=	1.675×10^{-27} kg
Avogadro number	N_A	=	6.022×10^{23} per mole
Universal gas constant	R	=	8.314 J mol^{-1} K^{-1}
Planck constant	h	=	6.63×10^{-34} J s
Speed of light	c	=	3.00×10^{8} m s^{-1}
1 atomic mass unit (amu)	1 amu	=	1.66×10^{-27} kg
	1MeV	=	9.648×10^{7} kJ mol^{-1}

The use of non-programmable calculator is permitted.

01. a) Write the IUPAC names of following compounds.

i. $[CoBr(NH_3)_5]SO_4$
ii. $[Co(SO_4)(NH_3)_5]^+$

b) Write down the molecular formulae of the following co-ordination compounds.

i. Ammonium tetrachloridocuprate(II)
ii. Sodium chloridopentacyanidoferrate(III)

c) Explain the following:

i. $[Ni(CN)_4]^{2-}$ is diamagnetic.

ii. $[NiCl_4]^{2-}$ is paramagnetic.

(Ni = 28)

d) Give four different types of ligands and give one example for each type.

e) Calculate the binding energy (in MeV) per nucleon for the isotope $^{56}_{26}Fe$.

(the mass of $^{56}_{26}Fe$ = 55.9349 amu).

(130 marks)

02. a) Draw the crystal field splitting diagram to explain the following observations:

i. $[Ni(CN)_6]^{2-}$, which has square planar geometry is diamagnetic.

ii. $[NiCl_4]^{2-}$, which has tetrahedral geometry is paramagnetic.

(30 marks)

b) Draw the octahedral crystal field splitting diagrams for Fe^{2+} with both weak and strong octahedral fields. Label the diagrams weak and strong field, high spin and low spin, give the names of the d-orbitals, and label the appropriate orbital sets e_g and t_{2g}.

(30 marks)

c) Find the spin only magnetic momentum of the octahedral complex ions of $[Fe(H_2O)_6]^{2+}$, $[Fe(CN)_6]^{4-}$ and $[Ni(NH_3)_6]^{2+}$ and deduce the magnetic properties of each of the above ions.

(40 marks)

03. a) Give two differences between amorphous and crystalline solids, name one example for each type of solid.

(30 marks)

b) Calculate the packing density or atomic packing factor of the following crystals structures and calculate the void percent in each structure.

i. Face-centered cubic structure.

ii. Body-centered cubic structure.

(40 marks)

c) i. Compute the Miller indices for a plane intersecting at x = ¼ , y =1, and z = ½ in a simple cubic lattice.

ii. Draw the plane and determine the axis intersects of a surface with the Miller index (013) in a simple cubic lattice.

(30 marks)

04. a) Explain the following terms;

i. Nuclear binding energy.
ii.Nuclear fission. (20 marks)

b) Write a balanced nuclear equation for each decay process indicated.
i. The isotope ^{234}Th decays by an alpha emission.
ii.The isotope ^{11}C decays by an electron capture.
(20 marks)

c) $^{24}_{11}Na$ is an unstable isotope of sodium with a half-life of 15 hours. Calculate the radioactive decay constant. State any assumption/s you have made in this calculation.
(30 marks)

d) Briefly discuss the use of radionuclides in agriculture and medicine.
(30 marks)

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